

RECONNAISSANCE: INFORMATION GATHERING FOR THE ETHICAL HACKER

Ethical Hacking and Penetration

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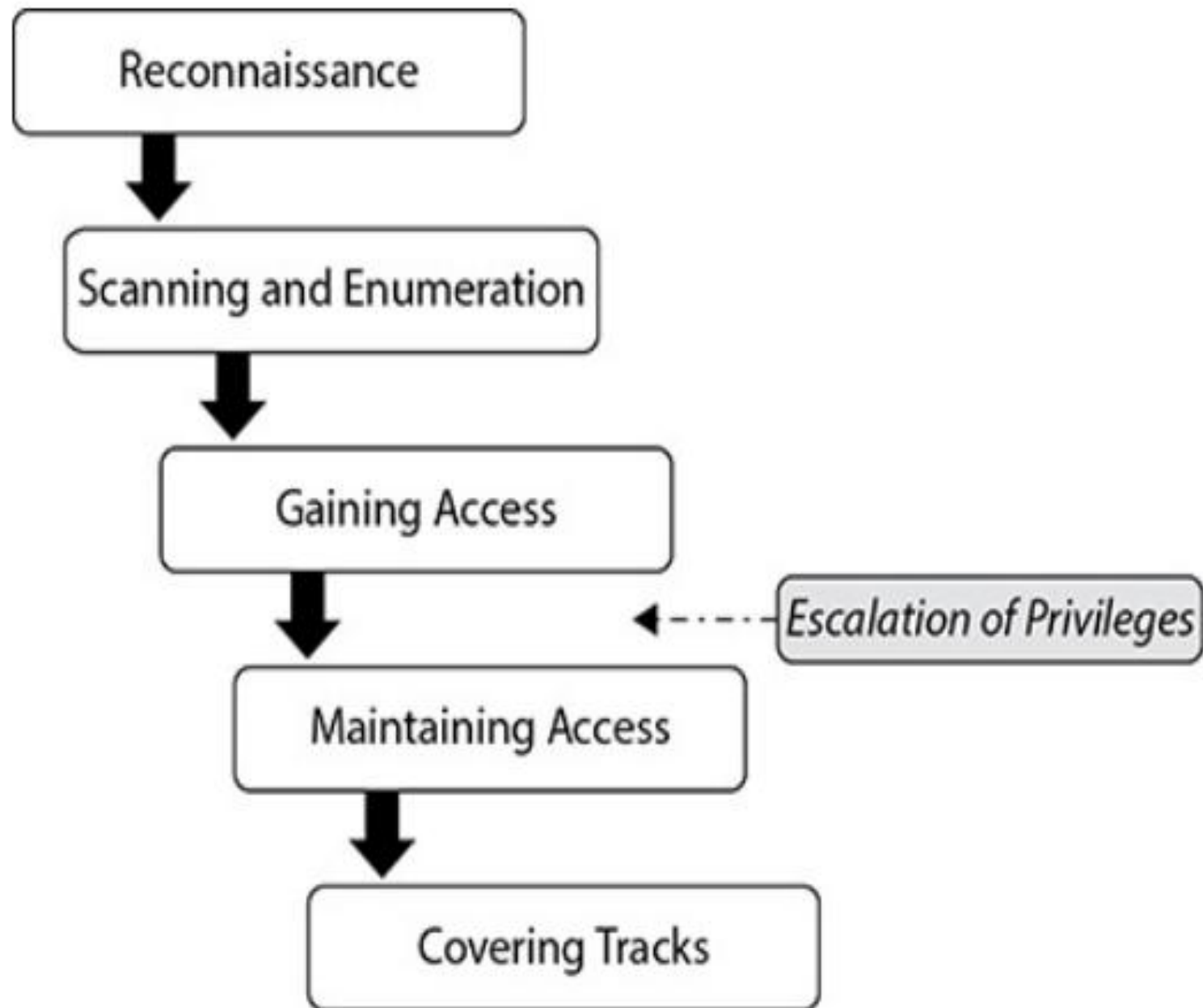
OUTLINE

- Active and passive footprinting
- Methods and procedures in information gathering
- The use of social networking, search engines, and Google hacking in information gathering
- The use of whois, ARIN, and nslookup in information gathering
- Describing the NS record types

INTRODUCTION

The first step in ethical hacking is reconnaissance, which means **gathering information about the target**. This process can be divided into two types: passive and active.

- Passive reconnaissance is the collection of information without directly interacting with the target, so the target does not know that any data has been gathered.
- Active reconnaissance involves direct interaction with the target's systems using tools or probes. While active methods often provide more detailed information, they also carry a greater chance of being detected, which puts the hacker's activities at higher risk.



FOOTPRINTING

FOOTPRINTING

Footprinting is more than just the first step of an attack; it is an essential skill for any ethical hacker. During this stage, the goal is to gather any information about the target, no matter how big or small. This information does not have to be technical. Useful data can include network architecture, public-facing websites, physical security measures, and even employee routines.

Information about employees is also very valuable, as employees represent a major target later in the test. Much of this data is relatively easy to obtain if one knows where to look.

FOOTPRINTING

As we mentioned, information gathering, or footprinting, can be performed using two primary approaches:

1. Passive footprinting
2. Active footprinting.

PASSIVE FOOTPRINTING

Passive footprinting involves collecting information from publicly available sources without directly interacting with the target system, network, or organization. Common sources include websites, social media platforms, public records, search engines, and online databases. Since no direct contact is made with the target, passive footprinting generally presents a low risk of detection.

Example:

Passive Footprinting: Examining an organization's website, reviewing employee profiles on social media, searching public DNS records, or collecting information from search engines.

ACTIVE FOOTPRINTING

Active footprinting involves direct interaction with the target environment to obtain additional information. This may include activities such as network scanning, port scanning, service enumeration, and other techniques that communicate with target systems. Although active footprinting often provides more detailed and accurate information, it increases the likelihood of detection by generating observable network traffic.

Example:

Active Footprinting: Performing a ping sweep, conducting a port scan with tools such as Nmap, or enumerating services running on target systems.

FOOTPRINTING METHODS AND TOOLS

FOOTPRINTING METHODS AND TOOLS

Search Engines

Google Hacking

Website Footprinting

Email Footprinting

DNS Footprinting

Network Footprinting

SEARCH ENGINES

SEARCH ENGINES

Search engines can reveal a wide range of information, including company websites, contact details, employee profiles, publicly accessible documents, subdomains, email addresses, and technology-related information. In addition, mapping services such as Google Maps and Bing Maps can provide geographical information about an organization's facilities, including building locations, surrounding infrastructure, and satellite imagery.

Because search engines and public information sources require no direct interaction with target systems, they are often the starting point of the reconnaissance process and can provide significant intelligence before active footprinting techniques are performed.

SEARCH ENGINES

Information Sought	Source
Company location	Google Maps
Employee names and positions	LinkedIn
Public documents (PDF, DOCX)	Google Search
Email addresses	Search engines, company website
Web server information	Netcraft
Domain ownership information	WHOIS databases

SEARCH ENGINES

Job Boards and Recruitment Websites

Job advertisement platforms are valuable sources of information during the footprinting phase of an ethical hacking assessment. Organizations frequently publish technical requirements when recruiting employees, and these requirements can unintentionally reveal details about the organization's technology infrastructure.

Job postings may disclose information about operating systems, database management systems, web technologies, cloud platforms, security solutions, programming languages, and network administration tools used within the organization. By analyzing these descriptions, security professionals can gain insight into the target environment and identify technologies that may be present on the network.

SEARCH ENGINES

Examples of information obtained from job postings:

- Operating systems (Windows Server, Linux, Ubuntu)
- Database systems (MySQL, PostgreSQL, Microsoft SQL Server)
- Cloud platforms (AWS, Azure, Google Cloud)
- Security tools (Splunk, Fortinet, Palo Alto)
- Virtualization technologies (VMware, Hyper-V)
- Programming languages and frameworks

SEARCH ENGINES

Social Media Intelligence (SOCMINT)

Social media platforms are important sources of publicly available information during reconnaissance activities. Employees often share professional achievements, job responsibilities, organizational activities, and technical expertise through social networking platforms. This information can help security professionals understand an organization's structure, technologies, and personnel.

Professional networking platforms such as LinkedIn may reveal employee roles, departmental structures, certifications, and technology stacks. Other social media platforms may provide information about company events, organizational changes, partnerships, and publicly disclosed projects.

GOOGLE HACKING

GOOGLE HACKING

Google Hacking Database (GHDB)

The Google Hacking Database (GHDB) is a publicly available collection of advanced Google search commands. These commands, often called "Google Dorks," are designed to find specific types of information that ordinary searches typically do not reveal.

<https://www.exploit-db.com/google-hacking-database>

GOOGLE HACKING

<https://www.exploit-db.com/google-hacking-database>

The screenshot shows the Exploit Database Google Hacking Database interface. The header features the Exploit Database logo and navigation icons. The main content area includes a search bar with filters for Type, Platform, Author, Port, and Tag. The search results are displayed in a table with columns for Date, D, A, V, Title, Type, Platform, and Author. The table lists several exploits, including WebExec - Authenticated User Code Execution (Metasploit), ServersCheck Monitoring Software 14.3.3 - Arbitrary File Write, NoMachine < 5.3.27 - Remote Code Execution, Delta Electronics Delta Industrial Automation COMMGR 1.08 - Stack Buffer Overflow (Metasploit), Microsoft Edge - Sandbox Escape, and Ubisoft Uplay Desktop Client 63.0.5699.0 - Remote Code Execution.

EXPLOIT DATABASE

Type: remote Clear Platform: Windows Clear Author: Begin typing... Port: Any Tag: Any Advanced

Verified Has App Filters Reset All

Show 15 Search:

Date	D	A	V	Title	Type	Platform	Author
2018-10-25	↓	✓		WebExec - Authenticated User Code Execution (Metasploit)	Remote	Windows	Metasploit
2018-10-23	↓	✗		ServersCheck Monitoring Software 14.3.3 - Arbitrary File Write	Remote	Windows	hyp3rlinx
2018-10-15	↓	✗		NoMachine < 5.3.27 - Remote Code Execution	Remote	Windows	hyp3rlinx
2018-10-09	↓	✓		Delta Electronics Delta Industrial Automation COMMGR 1.08 - Stack Buffer Overflow (Metasploit)	Remote	Windows	Metasploit
2018-09-27	↓	✓		Microsoft Edge - Sandbox Escape	Remote	Windows	Google Security Research
2018-09-18	↓	✗		Ubisoft Uplay Desktop Client 63.0.5699.0 - Remote Code Execution	Remote	Windows	Che-Chun Kuo
2018-09-12	↓	✗		Microsoft DirectX SDK - JXast-level Remote Code Execution	Remote	Windows	hyp3rlinx

GOOGLE HACKING

Google Hacking (Google Dorking)

Google Hacking, also known as Google Dorking, is a footprinting technique that uses advanced search operators to locate specific information indexed by search engines. Rather than performing ordinary keyword searches, specialized search operators are used to discover sensitive information, exposed resources, misconfigured systems, or publicly accessible files.

Information that may be discovered:

- Public documents
- Email addresses
- Login pages
- Directory listings
- Exposed configuration files
- Publicly accessible reports and presentations

GOOGLE HACKING

Operator	Example Query	Purpose
filetype	site:company.com filetype:pdf	Find PDF documents on a company's website.
index of	intitle:"index of" backup	Search for exposed backup directories.
info	info:company.com	View Google's information about a website.
intitle	intitle:"login"	Find login pages.
inurl	inurl:admin	Locate administrative interfaces.
link	link:company.com	Identify pages linking to a website.
related	related:microsoft.com	Find websites similar to Microsoft.
site	site:company.com	Display pages indexed from a specific website.

GOOGLE HACKING

Operator	Example Query	Purpose
related	related:cnn.com	Search for sites related to a given domain.
allintitle	allintitle:android google	Search for pages with multiple words in the title tag.
allinurl	allinurl:android google	Search for pages with multiple words in the URL.
intext	intext:apple	Search for pages with a particular word in their content.
allintext	allintext:apple iphone	Search for pages with multiple words in their content.
before	apple before:2022-12-29	Search for results from before a particular date.
after	apple after:2022-12-29	Search for results from after a particular date.

GOOGLE HACKING

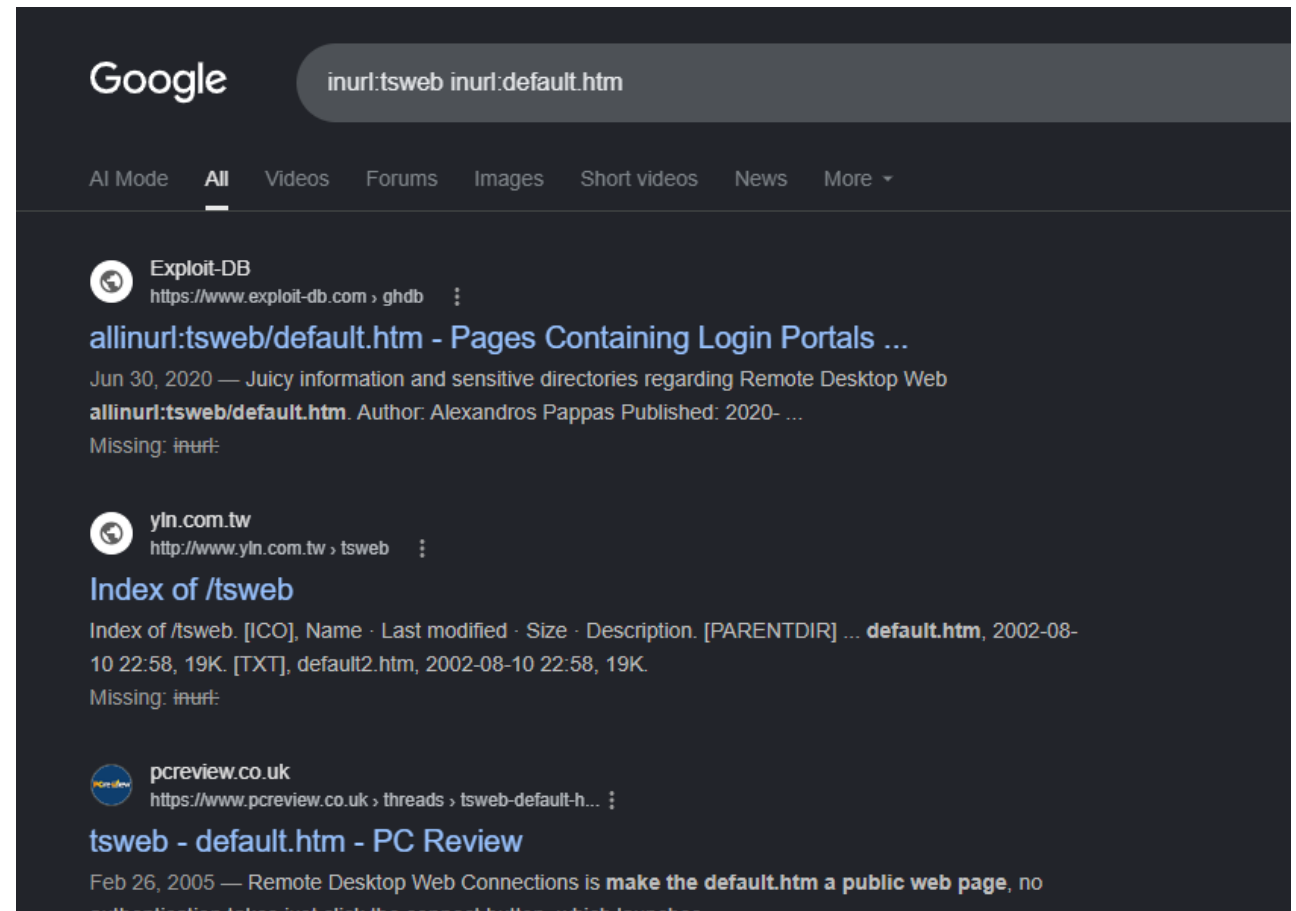
Operator	Syntax	Description
" " (Quotation Marks)	"admin login"	Searches for the exact phrase as written.
OR	admin OR login	Returns results containing either term.
AND	admin AND login	Returns results containing both terms. (Google usually assumes AND by default.)
- (Minus)	windows -linux	Excludes pages containing the specified term.
* (Wildcard)	"password * file"	Represents one or more unknown words within a search phrase.
()	(admin OR login) site:example.com	Groups search terms and operators to control the search logic.
.. (Range)	server 2016..2025	Searches for values within a specified numeric range.

GOOGLE HACKING

Examples:
using

`allinurl:tsweb/default.htm`

for finding web servers that have a
specific remote access tool



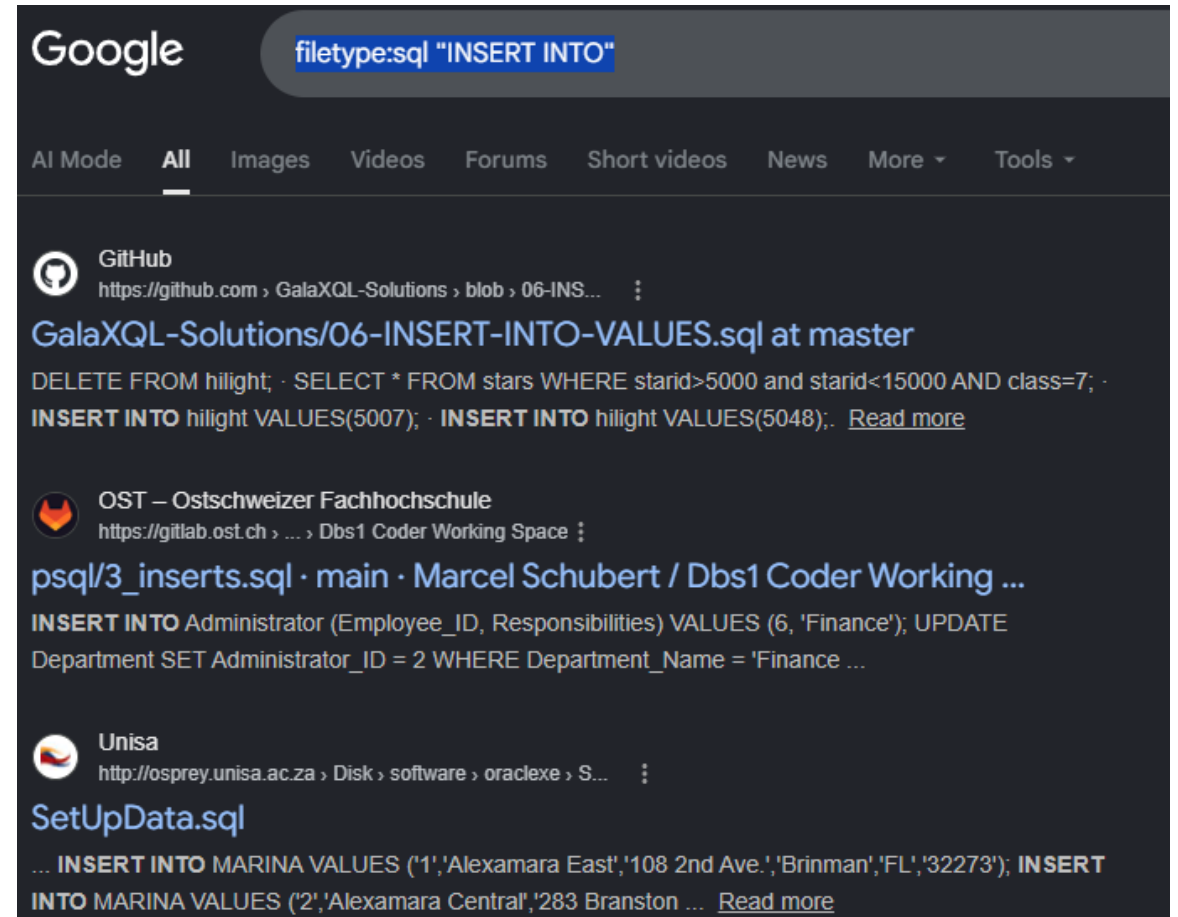
GOOGLE HACKING

Examples:
using

`filetype:sql "INSERT INTO"`

Exposes database backups/schemas

“SQL dumps are frequently left in web roots; contains usernames, emails, hashed passwords”



GOOGLE HACKING

Try also using the following Google Dorking :

`intitle:"index of /"`

`inurl:phpmyadmin/index.php`

`intitle:"admin login"`

`site:github.com "password" filetype:txt`

Explain the results.

SHODAN

Shodan (Sentient Hyper-Optimized Data Access Network) is a specialized search engine. Unlike Google, which searches for websites and documents, Shodan searches for devices connected to the internet.

These devices include:

- Computers and servers
- Routers and network equipment
- Webcams and security cameras
- Medical devices
- Industrial control systems (power plants, factories)
- Smart home devices (thermostats, doorbells, refrigerators)

SHODAN

https://www.shodan.io/

SHODAN

Explore

Downloads

Pricing

port:22

Account

27,747,553

TOP COUNTRIES

United States7,320,676

Brazil3,486,657

China2,453,981

Germany2,176,395

Argentina1,297,988

More...

TOP ORGANIZATIONS

TELEFÔNICA BRASIL S.A.3,123,330

Google LLC2,128,957

DigitalOcean, LLC1,367,213

Telefonica de Argentina1,244,991

Amazon Technologies Inc.906,623

More...

TOP PRODUCTS

OpenSSH18,759,417

Dropbear sshd4,745,301

View Report

Browse Images

View on Map

Advanced Search

Partner Spotlight: Looking for a Splunk alternative to store all the Shodan data? Check out [Gravwell](#)

8.210.183.170

Alibaba Cloud (Singapore) Private Limited

Hong Kong, Hong Kong

cloud

SSH-2.0-OpenSSH_7.4

Key type: ssh-rsa

Key: AAAAB3NzaC1yc2EAAAADAQABAAQCS0XJTDxYlHGbiXnkgXBR2Vh0u+frj7qC9LjqEaCzCusMDYekY0FLWqNw1qMTAs/TiW/oUMEVCA1SL+p80oXVp6LT14JrR3yJwBw4Jo8+JAnnMI1M7LEKcJ3Nxd/YUejK/SaJGh/qeTR0yHk7r1V5559aNTb9Xv70n+7yPUkwadVmJ4Kw450CkvzYmmq0hS+zHq7TF7qneu8+K8mmU3h1NQmIa5...

2024-09-03T03:11:16.805017

2a03:f480:1:d::8a

sed20ba00.fastvps-server.com

IPv6 network for hosting services

Estonia, Jõhvi

SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.10

Key type: ssh-rsa

Key: AAAAB3NzaC1yc2EAAAADAQABAAQGCQZbxt8FJTDpPciJxeZE09lHrH7ISRswcHfMQDnwZZbBu swrljUHM9rvyX/hJaZf2maw9yBZqsvA85EK0pfdJMM4m180ZbaqTnahqyFyouW2AxhUNtJ27CVfEWz0EN2yvgGnA5BwwYBkedlMca1Wh0gecEKmS1/juAH50AwHTp4FtDTm0vrcRcZBfEPb3AxtS+Dzsyd...

2024-09-03T03:11:15.899482

203.123.51.226

203-123-51-226.clgmail.com

Merce Broadside

Singapore, Singapore

SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.10

Key type: ecdsa-sha2-nistp256

Key: AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBB0YjuWKnHgLvqmQmpIOTWeG0ch7AigdfAG9P8Y+SePDxdef13byETQ3vLykKM0PAPWuWmp0//cEdaV2tllXrf4=Fingerprint: 2e:f2:0f:26:4c:6c:3d:61:2b:39:d1:f9:ab:b1:5b:f3

Kex Algorithms: ...

2024-09-03T03:11:14.762349

41.225.184.114

ATI - Agence Tunisienne Internet

Tunisia, Tunis

No data returned

2024-09-03T03:11:14.735588

29

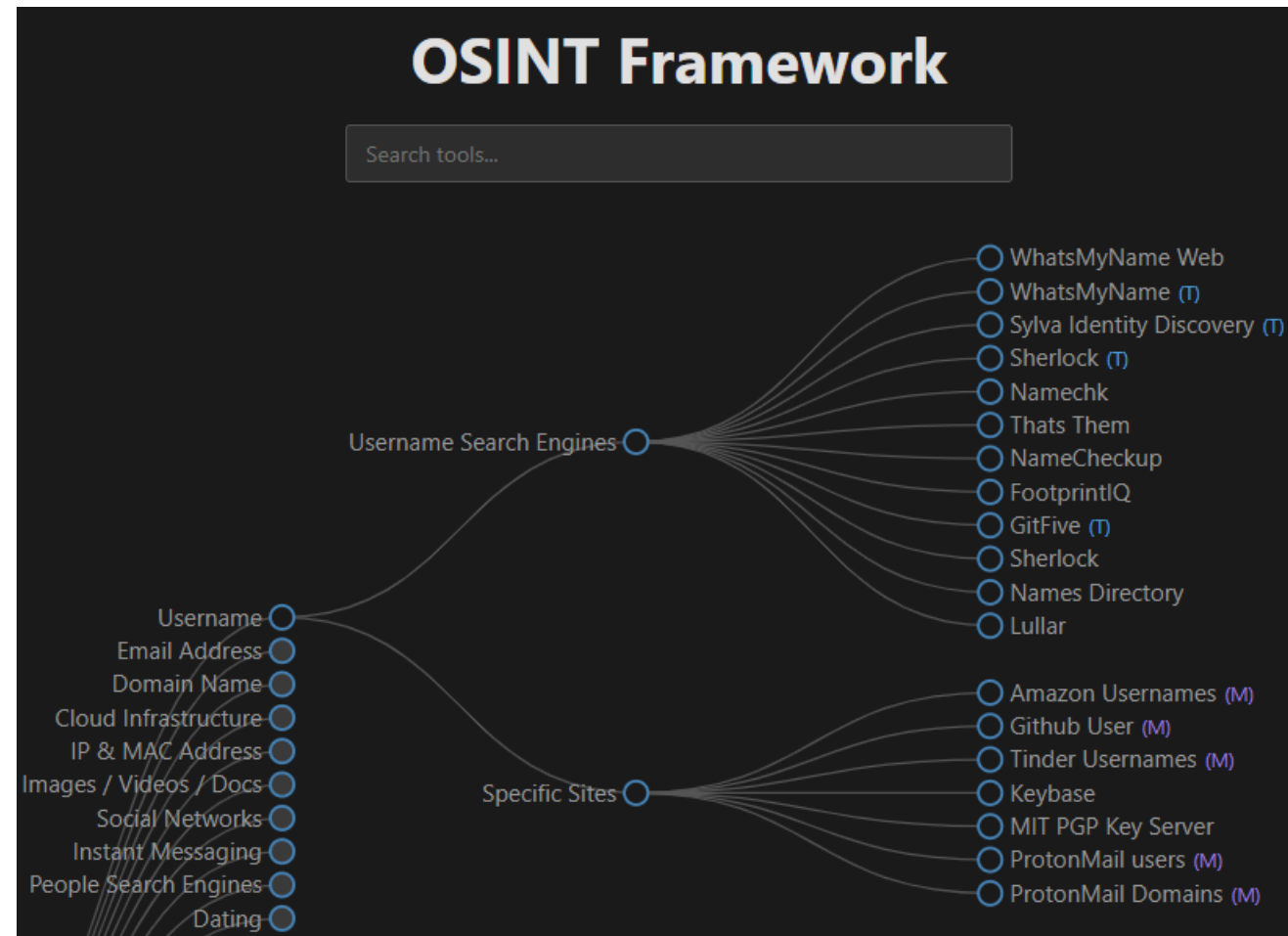
OSINT (OPEN-SOURCE INTELLIGENCE)

OSINT

Open-Source Intelligence (OSINT) is the practice of collecting, analyzing, and synthesizing publicly available information to produce actionable intelligence

OSINT tools may or may not be open source. Some tools are free and open, others require registration to use free versions, and others require a fee for use.

<https://osintframework.com>



Maigret Tool

Maigret is a powerful open-source OSINT (Open-Source Intelligence) tool for Kali Linux that automates the collection of a target's digital footprint by checking a username against thousands of sites. It is highly efficient for discovering linked accounts, unmasking users, and verifying identities.

[illegible]

WEBSITE FOOTPRINTING

WEBSITE FOOTPRINTING

Website footprinting is the process of collecting publicly available information about a target website. The goal is to identify technologies, services, and resources used by an organization without directly accessing internal systems. This information can help security professionals understand the target's online presence and infrastructure. Website footprinting is generally considered a passive reconnaissance technique.

Website analysis can reveal:

- Operating system information
- Web server details
- Software and applications in use
- File and directory names
- Contact information

WEBSITE FOOTPRINTING

The HTML source code of a website often contains useful information that is not immediately visible to users. Hidden form fields, metadata, and developer comments may reveal details about the website's structure and technologies. Cookies can also provide clues about web applications, frameworks, and scripting methods used by the organization. Developers occasionally leave comments in source code that disclose usernames, file paths, or configuration details.



WEB MIRRORING

Web mirroring is the process of creating a complete local copy of a website. Having an offline copy allows security professionals to explore the website structure in greater detail and perform analysis without repeatedly accessing the target website.

Common Web Mirroring Tools:

- HTTrack (www.httrack.com)
- Black Widow (<http://softbytelabs.com>)
- WebRipper (www.calluna-software.com)
- Teleport Pro (www.tenmax.com)
- GNU Wget (www.gnu.org)
- Backstreet Browser (<http://spadixbd.com>)



EMAIL FOOTPRINTING

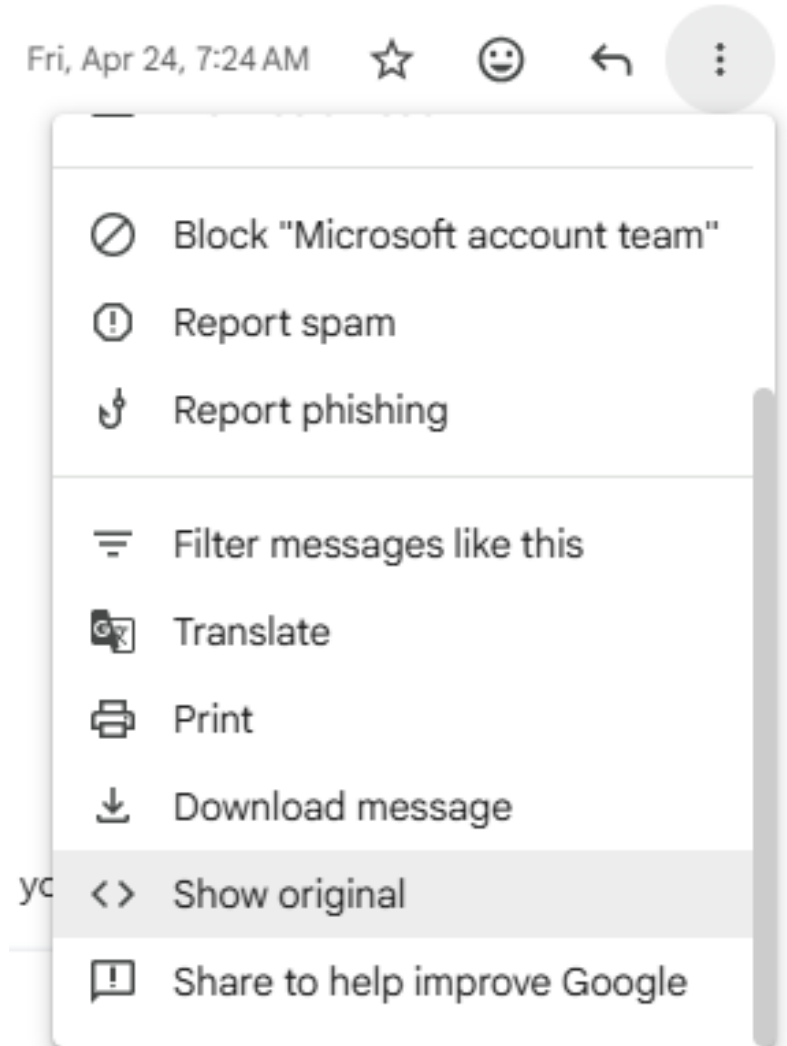
EMAIL FOOTPRINTING

Email communication can be an important source of information during the reconnaissance phase. Messages exchanged between individuals and organizations often contain technical details that may reveal information about systems, networks, and users.

By analyzing email-related information, security professionals can gain insights into the target organization's infrastructure and communication methods.

Email communications may reveal:

- IP addresses
- Geographic location information
- Email server details
- Domain names
- Communication patterns



EMAIL FOOTPRINTING

Email header

An email header contains technical information that describes how a message traveled from the sender to the recipient. Although hidden from normal view, headers can be displayed and analyzed using most email clients.

Email headers may contain:

- Sender information
- Recipient information
- Date and time stamps
- Mail server names
- Routing information
- Domain details

```
header.from=accountprotection.microsoft.com; dkim=none
(message not signed); arc=none
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed;
d=accountprotection.microsoft.com; s=selector1;
h=From:Date:Subject:Message-ID:Content-Type:MIME-Version:X-
MS-Exchange-SenderADCheck;
bh=L1qZX2Wnfd4QGG7oDUWjN0xZonVCcEQXnZXLcSRTqrk=;
b=eo6AmXBJbKdaZEK1ubbk9yiJg/Qq4/oIWZpZxNDR01wKURUROIc7XrwOeIc
yjic204y4H0BRAiej3ueyzITrD6LrvFwRQWedECej4LnFK5cFcjNU0a+HcWqm
FoYy0sWKWVi4wKX6ZzWpkPHuxNtWjUea86nXkzjQVywDTWSQnn4=
Received: from BL1PR13CA0335.namprd13.prod.outlook.com
(2603:10b6:208:2c6::10) by
MN0PR16MB6512.namprd16.prod.outlook.com
(2603:10b6:208:4cb::18) with Microsoft SMTP Server
(version=TLS1_2,
cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id
15.20.9846.22; Fri, 24 Apr 2026 04:24:39 +0000
Received: from BL02EPF0001A106.namprd05.prod.outlook.com
(2603:10b6:208:2c6:cafe::d9) by
BL1PR13CA0335.outlook.office365.com (2603:10b6:208:2c6::10)
with Microsoft SMTP Server (version=TLS1_3,
cipher=TLS_AES_256_GCM_SHA384) id 15.20.9846.20 via Frontend
Transport; Fri, 24 Apr 2026 04:24:38 +0000
X-MS-Exchange-Authentication-Results: spf=none (sender IP is
23.98.229.239) smtp.mailfrom=accountprotection.microsoft.com;
dkim=none (message not signed) header.d=none;dmarc=none
action=none header.from=accountprotection.microsoft.com;
Received: from accountprotection.microsoft.com
```

DNS FOOTPRINTING

DNS FOOTPRINTING

The Domain Name System (DNS) consists of servers distributed across the world. Each DNS server is responsible for managing records within a specific part of the DNS hierarchy, known as a **namespace**. These records provide information about different network resources and services.

DNS records may contain:

- IP addresses of hosts and servers
- E-mail server information
- References to other DNS servers
- Information that helps users locate network resources

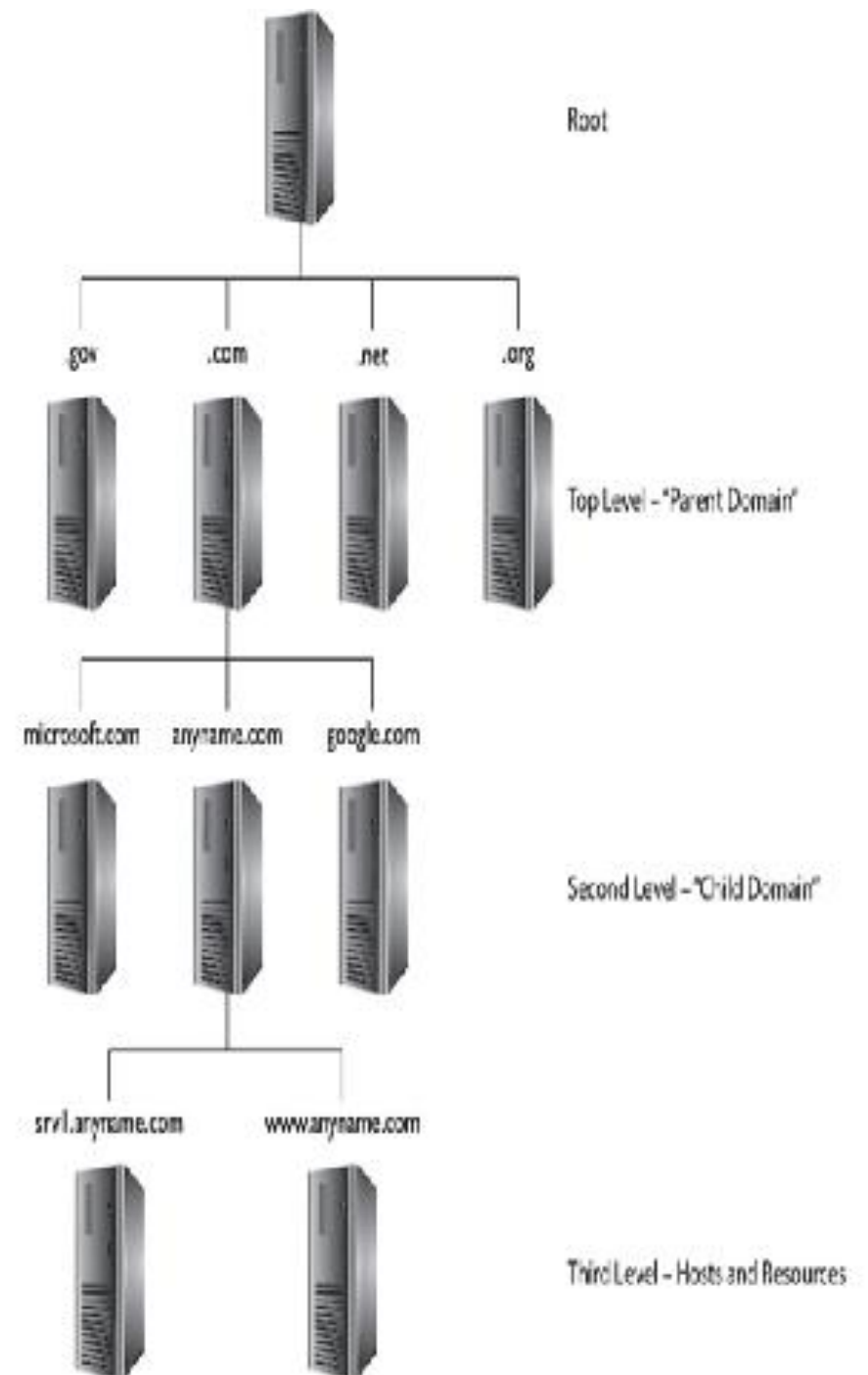
DNS FOOTPRINTING

DNS follows a hierarchical structure in which each server manages only its assigned namespace. Large DNS servers may manage top-level domains (TLDs) such as **.com**, while lower-level servers manage specific domains such as **mheducation.com**.

For example:

The DNS server responsible for **anyname.com** manages all records related to that domain.

Users searching for resources within that domain, such as a website or mail server, can obtain the required information from that DNS server.



DNS FOOTPRINTING

Although DNS is essential for Internet communication, its records can reveal valuable information about a network's structure.

By examining DNS records, an attacker may identify:

- The authoritative DNS servers
- E-mail servers used by the organization
- Public-facing web servers
- Other important network resources

As a result, DNS information can be a valuable source of intelligence during the footprinting and reconnaissance phases of an attack.

WHOIS

Whois is becoming ubiquitous in operating systems everywhere; it queries the registries and returns information, including domain ownership, addresses, locations, and phone numbers.

<https://www.whois.com/whois/>

[Domains](#)[Hosting](#)[Websites](#)[Email](#)[Security](#)[Whois](#)[Deals](#)

mhedcation.com

Updated 1 second ago



Domain Information

Domain:	mhedcation.com
Registered On:	2014-07-09
Expires On:	2027-07-09
Updated On:	2026-06-13
Status:	client transfer prohibited
Name Servers:	222.ns1.abovedomains.com 222.ns2.abovedomains.com



Registrar Information

Registrar:	Spaceship, Inc.
IANA ID:	3862

GEOLOCATION FOOTPRINTING

IP Address 23.98.229.239

Free IP lookup to check an IP address and search information like country, city, ISP, Proxy data and more.

IP Lookup Result

[Share The Result](#)

Geolocation Data

The geolocation data uses [IP2Location DB26](#) geolocation database.

- ☐ **Permalink** <https://www.ip2location.com/23.98.229.239>
- ☒ **IP Address** [23.98.229.239](#)
- ☐ **Country**  [United States of America \[US\]](#)
- ☐ **Region** Texas
- ☐ **City** San Antonio
- ☐ **Coordinates of City** 29.424120, -98.493730 (29°25'27"N 98°29'37"W)
- ☐ **ISP** Microsoft Corporation
- ☐ **Local Time** 13 Jun, 2026 10:16 PM (UTC -05:00)
- ☐ **Domain** microsoft.com

<https://www.ip2location.com/>

NETWORK FOOTPRINTING

NETWORK FOOTPRINTING

Network footprinting is the process of identifying and defining the IP address range used by a target organization. Determining the network range helps security professionals understand the target's network structure and reduces the time required for further reconnaissance activities.

Knowing the start and end points of an organization's IP address range makes it easier to identify active hosts, services, and network devices during later phases of an assessment.

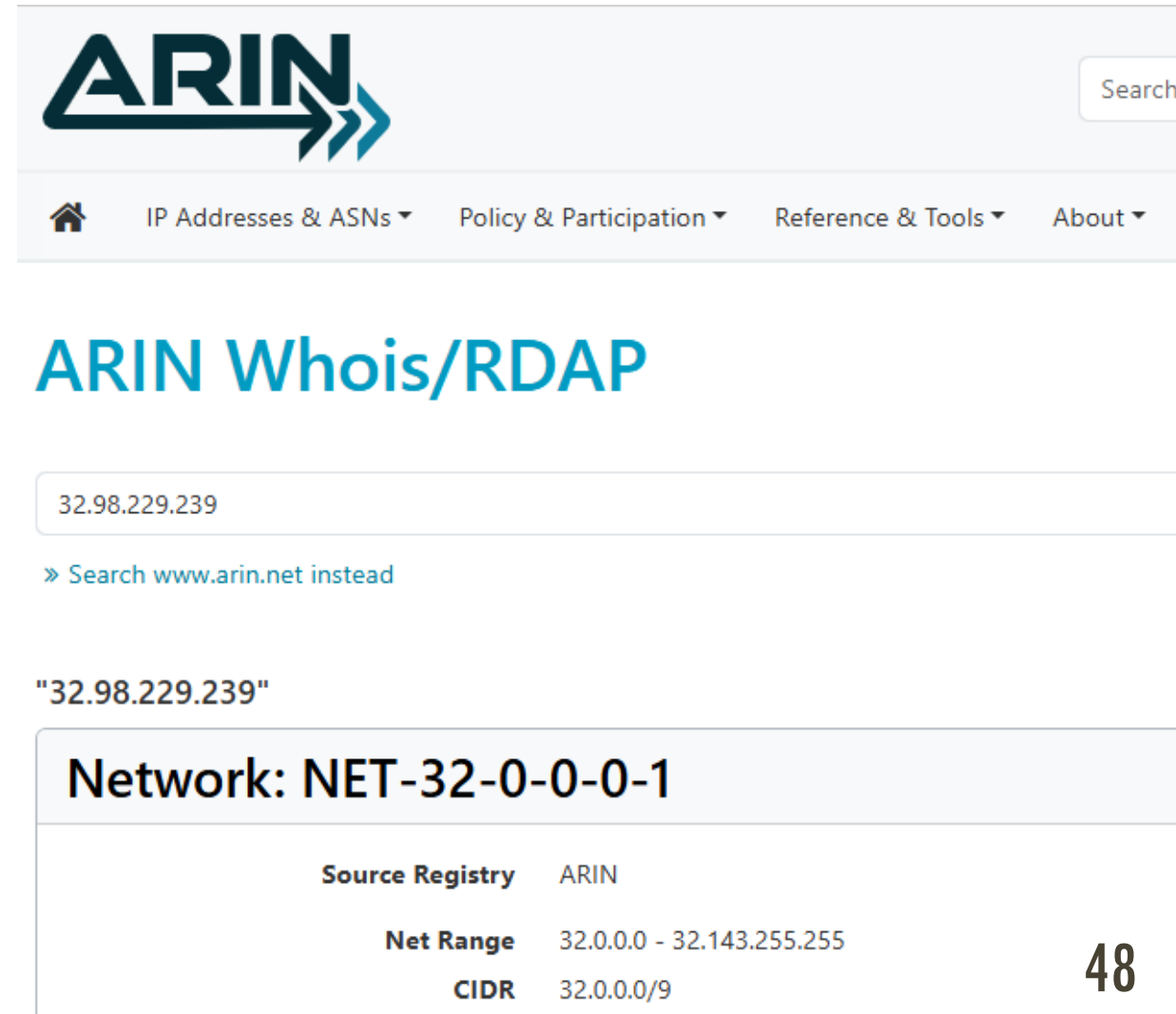
NETWORK FOOTPRINTING

ARIN as a Footprinting Resource

The **American Registry for Internet Numbers (ARIN)** maintains information about IP address allocations and network ownership.

By searching an IP address in ARIN, the following information may be obtained:

- Network address range
- Organization that owns the IP addresses
- Administrative contacts
- Technical contacts
- Additional registration details



The screenshot shows the ARIN website interface. At the top is the ARIN logo and a search bar. Below the logo is a navigation menu with links: Home, IP Addresses & ASNs, Policy & Participation, Reference & Tools, and About. The main heading is "ARIN Whois/RDAP". Below this is a search input field containing "32.98.229.239". A link below the input field says "» Search www.arin.net instead". Below the search field, the text "32.98.229.239" is displayed. Underneath, a box highlights the network "Network: NET-32-0-0-0-1". At the bottom, a table provides details about the network.

Source Registry	ARIN
Net Range	32.0.0.0 - 32.143.255.255
CIDR	32.0.0.0/9

NETWORK FOOTPRINTING

Traceroute for Network Mapping

Traceroute (or **tracert** in Windows) is a command-line tool used to track the path that packets take across a network from a source system to a destination system.

The tool helps security professionals identify the route followed by network traffic and measure the time required to reach each device along the path. This information is useful during the footprinting and reconnaissance phases of an assessment.

NETWORK FOOTPRINTING

Traceroute discovers the path to a destination by sending packets with gradually increasing **Time-To-Live (TTL)** values.

The process works as follows:

- A packet is sent with a small TTL value.
- Each router along the path decreases the TTL by one.
- When the TTL reaches zero, the router sends a response back to the source.
- Traceroute records the router's IP address, hostname, and response time.
- The TTL value is increased, and the process repeats until the destination is reached.
- This method reveals each network device, or **hop**, between the source and destination.

NETWORK FOOTPRINTING

Open the command prompt and type:
tracert example.com

```
C:\Windows\system32\CMD.exe X + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Zahra Rajeh>tracert google.com

Tracing route to google.com [172.217.168.142]
over a maximum of 30 hops:

  1    1 ms    1 ms    4 ms  192.168.1.1
  2   26 ms   27 ms   25 ms  82.114.163.111
  3   27 ms   26 ms   25 ms  82.114.160.6
  4   41 ms  191 ms  182 ms  82.114.164.1
  5  110 ms  128 ms  171 ms  185.200.127.8
  6  364 ms  357 ms  261 ms  213.202.4.175
  7  190 ms  182 ms  226 ms  216.239.41.83
  8  257 ms  373 ms  133 ms  192.178.105.70
  9  167 ms  307 ms  164 ms  108.170.249.116
 10  325 ms  144 ms  255 ms  192.178.105.97
 11  152 ms  167 ms  267 ms  192.178.120.249
 12  697 ms  657 ms  457 ms  172.253.71.228
 13  252 ms  259 ms  316 ms  142.251.243.250
 14  467 ms  500 ms   *      108.170.255.211
 15  204 ms  181 ms  220 ms  108.170.233.57
 16  288 ms  373 ms  398 ms  tzmila-af-in-f14.1e100.net [172.217.168.142]

Trace complete.
```

OTHER FOOTPRINTING TOOLS

OSRFRAMEWORK

OSRFramework

Purpose: Perform Open-Source Intelligence (OSINT) investigations.

Information Obtained:

- Usernames
- E-mail addresses
- Domains
- Phone numbers
- Online profiles

Use in Footprinting:

Automates information gathering from multiple public sources

WEB SPIDERS

Purpose: Crawl websites and collect information.

Information Obtained:

- Web pages
- Directories
- Links
- Publicly accessible resources

Use in Footprinting:

Maps website structure and identifies hidden or forgotten content.

MALTEGO

Purpose: Analyze and visualize relationships between entities.

Information Obtained:

- Domain relationships
- E-mail addresses
- Social media profiles
- Organizations and individuals

Use in Footprinting:

Helps correlate information and discover connections that may not be immediately visible.

Regardless of which methods you choose to employ, Footprinting is probably the most important phase of hacking you'll need to master. Spending time in this step drastically increases the odds of success later and is well worth the effort. Just maintain an organized approach and document what you discover.

